application (Ballot-Dapp) is used to illustrate Truffle-based development steps and deployment on a local Ganache test chain.

Part 2 (chapters 5-8) is about end-to-end Dapp development, with additional blockchain-specific features such as on-chain data, security, and privacy.

Chapter 5 introduces security and privacy in the context of blockchain programming. Cryptography and hashing algorithms and techniques are discussed at a high level. The concepts are illustrated by means of a blind auction smart contract (BlindAuction.sol).

Chapter 6 introduces the concept of on-chain and off-chain data, which is unique to blockchain programming. The blind auction and ASK smart contracts are extended into Dapps (BA-Dapp, ASK-Dapp) to demonstrate development with on-chain and off-chain data. Defining, emitting, and accessing blockchain events and logs are illustrated.

Chapter 7 focuses on the web3 API of Ethereum that enables web applications to access the underlying blockchain services. The blockchain side-channel concept is introduced to illustrate the use of web3 in a micropayment channel (MPC) application for massive plastics cleanup (MPC-Dapp).

Chapter 8 discusses deploying the smart contracts developed on a public cloudlike infrastructure called Infura. Infura is a web3 provider, and a gateway to public blockchains such as Ropsten (mainnet and IPFS). Public